Reg. No. | | | | | | | | | |

Question Paper Code : 16365

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2020

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. Information Technology

19CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A,CO1 1. The Karnaugh map for a Boolean function is given as

AB \ CD	C`D`	C`D	CD	CD`
A`B`	0	0	0	0
A`B	0	0	1	0
AB	1	1	1	1
AB`	0	1	1	1

The simplified Boolean equation for the above Karnaugh Map is

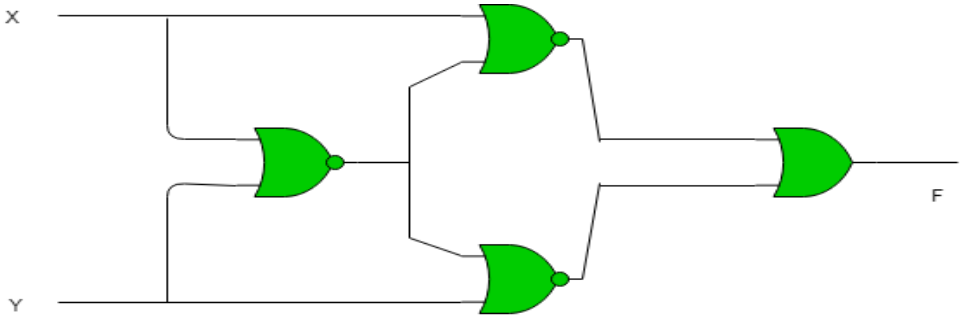
A) AB + CD + A`B + AD B) AB + AC + AD + BCD

C) AB + AD + BC + ACD D) AB + AC + BC + BCD

A, CO1 2. The Boolean function x'y' + xy + x'y is equivalent to

A) x' + y' B) x + y C) x' + y D) x + y'

A, CO2 3. Which of the following logic operations is performed by the following given combinational circuit?



A) EXCLUSIVE OR B) EXCLUSIVE NOR

C) NAND D) NOR

U, CO2 4. Draw the truth table of half adder.

A, CO3 5. How many 16K * 4 RAMs are required to achieve a memory with a capacity of 64K and a word length of 8 bits? Justify your answer.

A) 2 B) 4 C) 6 D) 8

U, CO3 6. Compare PAL and PLA.

A, CO4 7. The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0, 0, 1, 1, 2, 2, 3, 3, 0, 0,...) is _____. Justify.

A) 0 B) 1 C) 2 D) 3

A, CO4 8. How many states are there in a 3-bit ring counter? List them.

U, CO5 9. Differentiate synchronous and asynchronous sequential circuits.

R, CO5 10. Define static '1' and static '0' hazard.

PART B – (5 × 16 = 80 Marks)

A, CO1 11. a) Simplify the following switching functions using Karnaugh map method and realize expression using gates

A) F(A,B,C,D)=∑(0,3,5,7,8,9,10,12,15)

B) F(V, W, X, Y, Z)= Σ(0,2,6,10,12,16,20,25,27,29,30) (16 Marks)

OR

A, CO1 11. b) Simplify the following Boolean function using Quine-McClusky method.

F(A,B,C,D,E,F)=∑(0,5,7,8,9,12,13,23,24,25,28,29,37,40,42,44, 46, 55,56,57,60,61) and implement the circuit using NOR gates. (16 Marks)

A,CO2	12. a) i.	Implement the following Boolean function using 8 to 1 Multiplexer $F(A,B,C,D)=A^1BD^1+ACD+B^1CD+A^1C^1D$	(8 Marks)
A,CO2	12. a) ii.	Construct a 6 x 64 decoder using 3 x 8 decoders.	(8 Marks)
		OR	
A,CO2	12. b) i.	Design of four bit binary adder – subtractor circuit.	(6 Marks)
A,CO2	12. b) ii.	Design a combinational circuit that accepts a 3 bit number and generates a 6bit binary number output equal to the square of the input number. Write a high- level behaviour VHDL description for the circuit.	(10 Marks)
A, CO3	13. a) i.	Implement the Gray to Binary code convertor using ROM. Also, specify the size of ROM required to build a 4 bit binary adder.	(8 Marks)
U,CO3	13. a) ii.	Illustrate and describe the various categories of RAM with neat diagram.	(8 Marks)
		OR	
A,CO3	13. b) i	Implement the following two Boolean functions using PLA with 3 inputs, 4 product terms and 2 outputs. $F1=\sum m(3,5,6,7)$ $F2=\sum m(1,2,3,4)$	(8 Marks)
U,CO3	13. b) ii	Write notes on memory decoding.	(8 Marks)
A,CO4	14. a) i	Design a 3 bit binary up counter using T flip-flop.	(8 Marks)
U,CO4	14. a) ii	Construct a D-flip-flop using J-K flip-flop and T flip-flop.	(8 Marks)
		OR	
A,CO4	14. b) i.	Construct a mod 10 asynchronous counter using J-K flip-flop.	(8 Marks)
U,CO4	14. b) ii.	Draw and explain the parallel in serial out shift register and explain.	(8 Marks)
A,CO5	15. a) i.	An asynchronous sequential circuit is described by the following excitation and output function. $Y=X_1X_2+(X_1+X_2)\,Y$ $Z=Y$ A) Draw the logic diagram of the circuit. B) Derive the transition table and output map. C) Describe the behaviour of the circuit.	(8 Marks)

U,CO5	15. a) ii.	Explain different types of hazards in combinational and sequential circuits.	(8 Marks)

OR

A,CO5	15. b) i	Construct a static hazard free asynchronous circuit for the following switching function F=∑(1,3,5,7,8,9,14,15).	(8 Marks)
U,CO5	15. b) ii	Elaborate the process of reducing flow table while designing an asynchronous sequential circuit.	(8 Marks)